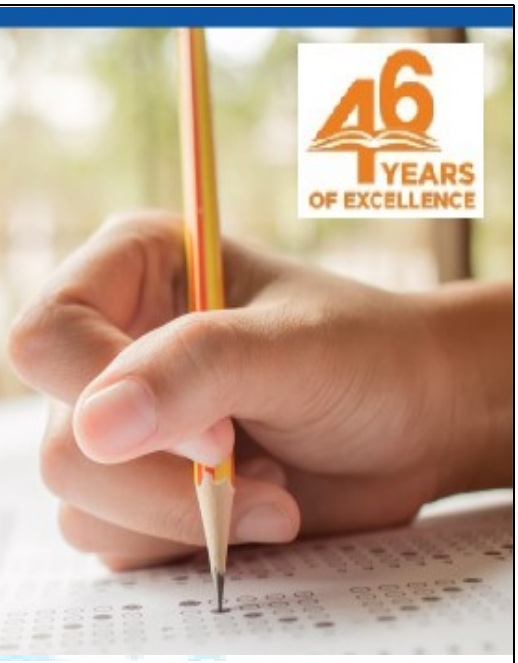
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 30-04-26
Time: 3 Hrs Max. Marks: 180

30-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-10(P1)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: TOTAL SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

THE NARAYANA GROUP

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 5 – 10)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 11 – 16)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 23)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 24 – 29)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 30 – 35)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 42)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 43 – 48)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 49 – 54)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. In quadrilateral $ABCD, BC = CD = AD = 10$ and $AB \| DC$. Then maximum possible area (in square units) of the quadrilateral is

A) 100 B) 75 C) $100\sqrt{3}$ D) $75\sqrt{3}$

2. Let $I = \int \frac{25e^{x/2}\cos x dx}{(3\cos x + 4\sin x)^{1/3}} = f(x) + C$ and $f(0) = 18$. Then maximum value of $f(x) \cdot e^{-x/2}$ is

A) 25 B) 5 C) 150 D) None of these

3. Value of $\int_{-2}^{2} \frac{\ln(2-x)}{\sqrt{4-x^2}} dx$ is

A) 0 B) $-\frac{\pi}{2}\ln 2$ C) $\pi \ell n 2$ D) None of these

4. If α and β are the roots of the equation $2z^2 - 3z - 2i = 0$, where $i = \sqrt{-1}$, then $\text{Re}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) \cdot \text{Im}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right)$ is equal to

A) 398 B) 312 C) 409 D) 441

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 5 and 6:

A student can solve 2 out of 4 problems of mathematics, 3 out of 5 problem of physics and 4 out of 5 problems of chemistry. There are equal number of books of math, physics and chemistry in his shelf. He selects one book randomly and attempts 10 problems from it.

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5. What is the probability that he will solve the first problem?

6. If he solves the first problem, then what is the probability that he will be able to solve the second problem as well?

Question Stem for Question Nos. 7 and 8:

$f(x)=x^n+a_1x^{n-1}+a_2x^{n-2}+....+a_{n-1}x+a_n$, where $a_i=\pm1$ for all $1\le i\le n$. Also $f(x)=0$ has all real roots.

7. How many distinct $f(x)$ can be possible

8. Maximum degree of $f(x)$ can be

Question Stem for Question Nos. 9 and 10:

A circle whose centre lies on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ intersects the rectangular hyperbola $xy=c^2$ in four points.

9. Eccentricity of locus of centroid of triangle formed by any of the three points is :

10. If locus of centre of locus of centroid of triangle formed by any of three point is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=K^2$ then $|K|$ is ____

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

11. Let A,B,C,D are 4 points on sphere of radius 1 unit and centre at O, such that $AB\bullet AC\bullet AD\bullet BC\bullet BD\bullet CD=\frac{2^9}{3^3}$. Which of the following is/are true?

A) $AB=\frac{2\sqrt{2}}{\sqrt{3}}$

B) Angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is $\cos^{-1}\left(\frac{-1}{3}\right)$

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C) Volume of tetrahedron ABCD is $\frac{8\sqrt{3}}{27}$

D) Area of ΔABC is $\frac{2\sqrt{3}}{3}$

12. The AB and CD are two perpendicular focal chords of a parabola $y^2 = 28x$. Which of the following statements are true for quadrilateral ACBD?
(Where A and C are on the same side of axis of parabola)
A) Minimum area is 1568 square units
B) Minimum area is 1538 square units
C) When area is minimum then length of each diagonal is 56 units
D) Sides AC and DB intersects on line $x + 7 = 0$

13. If $y = f(x)$ be a curve passing through (e, e^e) and which satisfy the differential equation $(2ny + xy\ell nx)dx - x\ell nxdy = 0$. Which of the following is/are true ? (Where $n \in N$)
A) $f(1) = 0$
B) If $h(x) = f(x)e^{-x}$, then $h(x) = h\left(\frac{1}{x}\right)$
C) If $g(x) = \lim_{n\to\infty} f(x)$, then $\int_{\frac{1}{e}}^{e} g(x)dx = 0$
D) $f(1) = e$

14. Let A and B be 2×2 matricies with real entries satisfying $(AB - BA)^n = I$ for some positive integer n. Possible values of n can be
A) 3 B) 5 C) 4 D) 8

15. Consider a pyramid OPQRSTU such that its base OPQRST is a regular hexagon of side length 2 units in xy plane with O as origin and P and S are along the +ve x -axis and +ve y -axis respectively. The point U is directly above the centre C of the hexagon such that $CU = \sqrt{5}$. Then
A) The acute angle between OP and OU is $\cos^{-1}\frac{1}{3}$
B) Equation of the plane OUS is $\sqrt{5}x - y = 0$
C) Shortest distance between lines OU and PQ is $\sqrt{3}$
D) Volume of the pyramid is $\sqrt{60}$ cubic units

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16. Let ℓ_1 and ℓ_2 be the lines $\vec{r}=\lambda\left(\hat{i}+\hat{j}+\hat{k}\right)$ and $\vec{r}=\left(\hat{j}-\hat{k}\right)+\mu\left(\hat{i}+\hat{k}\right)$, respectively. Let X. be the set of all the planes H that contain the line ℓ_1. For a plane H, let $d(H)$ denote the smallest possible distance between the points of ℓ_2 and H. Let H_0 be a plane in X for which $d(H_0)$ is the maximum value of $d(H)$ as H varies over all planes in X. then

A) The value of $d(H_0)$ is $\frac{1}{\sqrt{2}}$

B) The distance of the point $(0,1,2)$ from H_0 is $\sqrt{2}$

C) The distance of origin from H_0 is $\sqrt{2}$

D) The distance of origin from the point of intersection of planes $y=z, x=1$ and H_0 is $\sqrt{3}$

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

17. $0<a<b$, then the maximum value of $\left[8\frac{\ln\frac{b^2+4}{a^2+4}}{(b-a)}\right]$ is___([.] represents greatest integer function)

18. Number of possible integral values of that $f(x)=x^3+bx^2+3x+\sin x$ is bijective.

19. The square of the distance of the point $(-2,-8,6)$ from the line $\frac{x-1}{1}=\frac{y-1}{2}=\frac{z}{-1}$ along the line $\frac{x+5}{1}=\frac{y+5}{-1}=\frac{z}{2}$ is equal to

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PHYSICS **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

20. Consider a linear array of equally spaced dipoles, extending from minus to plus infinity. Assume that the positions of the dipoles are fixed and that each dipole, with moment of magnitude p, is free to point in any direction, as if it were mounted on a fully swiveling pivot. Of course, if we have an infinite array, the total energy may well be infinite, but we are interested in the energy per unit of the array. Compare the following arrangements in respect to average energy per dipole. Which has the highest energy?

A) 

B) 

C) 

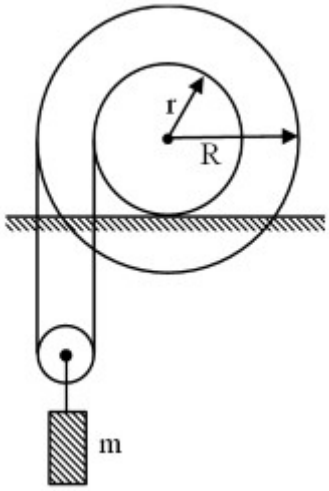
D) 

21. Two metallic strings A and B of different materials are connected in series forming a joint. The strings have similar cross-sectional area. The length of A is $\ell_A = 0.3$ m and that of B is $\ell_B = 0.5$ m. One end of the combined string is tied with a support rigidly and the other end is loaded with a block of mass 0.9 kg passing over a frictionless pulley. Transverse waves are set up in the combined string using an external source of variable frequency, calculate the lowest frequency for which standing waves are observed such that the joint is a node. The linear densities of A and B are 10 g / m and 2.5 g / m respectively.

A) 50 Hz B) 60 Hz C) 300 Hz D) 150 Hz

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22. A wheel with two rims (outer radius R , inner radius r) rolls with acceleration under the action of a load of mass m . Find acceleration of the block if the mass of the wheel is M and its moment of inertia about the centre of mass of the wheel is J .



A) $\frac{g}{\frac{4\left(J+Mr^2\right)}{m(R+r)^2}+1}$ B) $\frac{g}{\frac{2\left(J+Mr^2\right)}{m(R+r)^2}+1}$ C) $\frac{2g}{\frac{4\left(J+Mr^2\right)}{m(R+r)^2}+1}$ D) None of the above

23. A perfectly absorbing, black, solid sphere with constant density and radius R , hovers stationary above the sun. This is because the gravitational attraction of the sun is balanced by the pressure due to the sun's light. Assume the sun is far enough away that it closely approximates a point source of light. The distance from the centre of the sun at which the sphere hovers is:-

A) proportional to R B) proportional to $1/R^2$

C) proportional to $1/R$ D) independent of R

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 24 and 25:

On October 25, 2023, a new permanent exhibition space dedicated to the figure of Santiago Ramon y Cajal, Nobel Prize in Medicine in 1906, was inaugurated in the Paraninfo building of the University of Zaragoza, in recognition of his work on the structure of the nervous system. Through the axon, the nerve impulse is transmitted from the cell body to the axon terminals. We can consider the axon as a cylinder surrounded by the cell membrane and filled with axoplasm, which is a solution of

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various ions in water, among which Na^+ and K^+ stand out. The exterior of the axon membrane is also surrounded by a solution with different concentrations of the same ions. In the membrane there is a set of channels that allow or block the passage of ions, and some proteins called sodium-potassium pumps, which expel sodium and introduce potassium, at the cost of metabolic energy consumption. When the neuron is at rest, the inner wall of the cell membrane is negatively charged and the outer wall is positively charged, with an electrostatic potential difference $V_0 = -70mV$ with respect to the exterior of the membrane. In this resting situation, the concentration of Na^+ and K^+ ions inside and outside the membrane is as shown in Table.

Table. Concentration (mol/m^3)

Ion	Interior	Exterior
Na^+	15	145
K^+	150	5

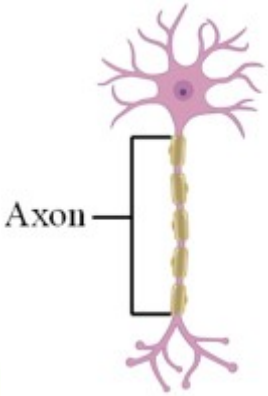
In addition to the effect of the electrostatic potential on the ions, the differences in concentration of each ion between the interior and the exterior tend to equalize through the membrane. This effect can be expressed by means of a potential V_x to which ion X is subjected, called the Nernst potential, which is determined by

$$V_x = \frac{RT}{eN_A} \ln\left(\frac{|X|_{ext}}{|X|_{int}}\right) \ldots..(1)$$

where R is the ideal gas constant, T is the temperature, e the elementary charge, N_A Avogadro's number, and $|X|_{int}$ and $|X|_{ext}$ the concentrations of ion X inside and outside the cell membrane, respectively. The "duel" between the electric interaction and the concentration gradient of both ions between the interior and the exterior of the cell results in a potential difference V_x eff , called the effective potential, different for each ion X, given by $V_x^{eff} = V_x - V_0 \ldots\ldots..(2)$

Consider that a K^+ ion passes from the interior to the exterior of the membrane without experiencing any type of friction, starting from rest at a point on the axis of the axon. Take $\ell n10 = 2.3, \ell n3 = 1.1$

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Data:

- Ideal gas constant : $R = 25/3 \text{ J mol}^{-1} \text{ K}^{-1}$
- Human body temperature : $T_{body} = 37^{\circ}\text{C}$
- Axon diameter: $D_A = 10\mu \text{ m}$
- Atomic masses: $m(\text{Na}) = 23\text{u}, m(\text{K}) = 39\text{u}$
- Atomic mass unit : $1\text{u} = 1.66 \times 10^{-27} \text{ kg}$
- Axon length : $L_a = 5 \text{ mm}$
- $N_A = 6 \times 10^{23} \text{ mol}^{-1}$

24. Determine the velocity of the K^+ ion (in m/s) upon leaving the cell membrane.

25. How long (in ns) does the K^+ ion take to travel the radius of the axon? (Approximate the motion to be uniformly accelerated.)

Question Stem for Question Nos. 26 and 27:

In this problem we model heat transfer in a simplified model of a honeycomb, shown in plan view in the figure. In figure-1, a bee (black circle in the center of the picture) is surrounded by a single layer of empty cells (colored light gray for clarity), whereas in figure-2 the bee is surrounded by a double layer of empty cells (for better clarity, the cells of the inner layer are colored light gray). Each vertical wall between two cells is a rectangle with a width a and height of b ; the wall thickness is d. Although a significant part of the thermal insulation is provided by the air in each cell between the bee and the outside, in this problem we assume that each cell has a uniform temperature and that heat is transferred only through the walls of the cells. The thermal conductivity of the walls in the model is λ. The layer of wax that closes the honeycomb below and above the cells is much thicker than the walls between the cells, so in the model we assume that heat is not transferred through the bottom and the top; heat is transferred only through the vertical walls of the cells. Each individual bee releases a heat power of P_0 and the outside temperature is T_0.

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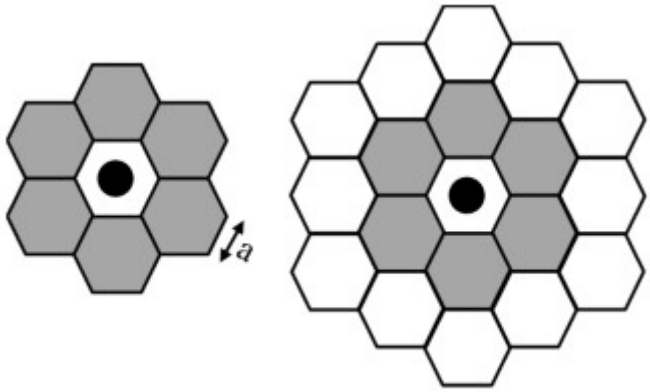


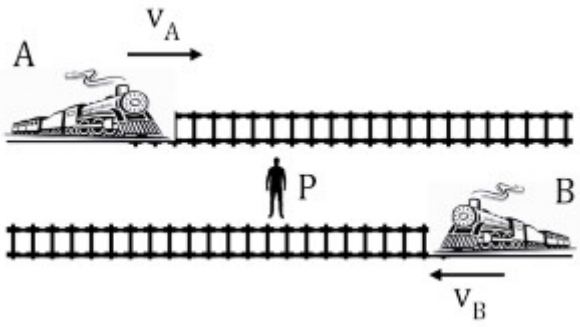
Figure-1 Figure-2

26. The temperature of the cell with the bee in the model with a single layer of cells (figure-1) is $T_0 + \frac{\alpha dP_0}{ab\lambda}$. The value of α is :-

27. The temperature of the cell with the bee in the model with a double layer of cells (figure-2) is $T_0 + \frac{\beta dP_0}{ab\lambda}$. The value of β is :-

Question Stem for Question Nos. 28 and 29:

As shown in Figure, an observer P is standing between two close parallel train tracks when two trains approach from opposite directions. Locomotive A has a speed $|v_A| = 15\ m/s$. It toots its whistle, which has a frequency $v_0 = 200\ Hz$. Locomotive B has a speed $|v_B| = 30\ m/s$. The speed of sound in the air is $340\ m/s$, and no breeze is blowing.



28. Find the wavelength λ_1 (in m) of the sound waves observer P receives from locomotive A.

29. Suppose that some of the sound waves reaching locomotive B are reflected back toward observer P and locomotive A. Find the wavelength λ (in m), of the reflected sound waves that observer P hears.

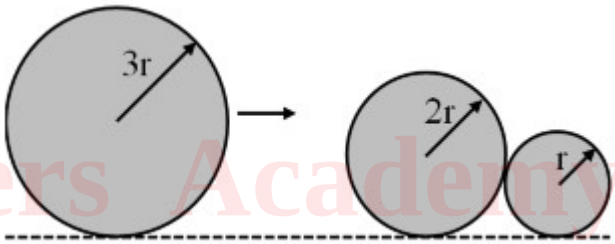
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SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

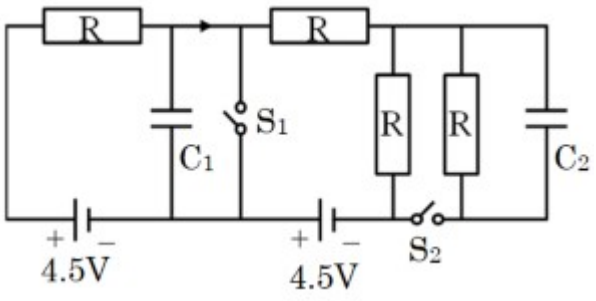
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

30. There are two discs of radii $2r$ and r at rest on a horizontal tabletop such that they almost touch each other. In the plane, a third disk of radius 3 r slides without rotation so that its velocity vector is parallel to a common tangent of the three disks. The top view of the three disks is shown in the figure. All three disks are made of the same material and have the same height. After the collision, which is considered to be elastic, the disc with radius 3 r slows down due to friction and stops after travelling $d = 5$ cm.



A) The disk of radius 2 r moves a distance 19 cm (approx.) before coming to rest.
B) The disk of radius r moves a distance 127 cm (approx.) before coming to rest.
C) The disk of radius 2 r moves in the direction $\tan^{-}(0.11)$ with the original direction of motion of disk of radius 3 r.
D) The disk of radius r moves in the direction $\tan^{-}(0.35)$ with the original direction of motion of disk of radius 3r.

31. In the circuit we connect four identical resistors $R = 300\Omega$, two switches S_1 and S_2, two identical capacitors $C_1 = C_2 = 4\mu$ F and two sources with an electromotive voltage of 4.5 V, as shown in the circuit diagram. Switch S_1 is kept open and switch S_2 is kept closed for a long time.



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A) The charge on capacitor C_1 is $3.6\mu C$

B) The magnitude of the current, immediately after we open switch S_2 at $t=0$ through the resistor that is farthest to the right in the diagram is 12 mA .

C) At $t=0$, we close switch S_1 then the magnitude of the current through switch S_1 after a long time is 5 mA .

D) At $t=0$, we close switch S_1 then the charge on capacitor C_2 after a long time is $6\mu C$

32. A sample of Thorium-232 $\left({}^{232}_{90}\text{Th}\right)$ undergoes alpha decay to form an isotope of Radium-228 $\left({}^{228}_{88}\text{Ra}\right)$. The reaction can be written as :

${}^{232}_{90}\text{Th} \rightarrow {}^{228}_{88}\text{Ra}^{*} + {}^{4}_{2}\text{He}$

Then, Radium-228 undergoes beta decay :

${}^{228}_{88}\text{Ra} \rightarrow {}^{228}_{89}\text{Ac}^{*} + e^{-1} + \bar{\nu}$

Both Radium-228 and Actinium-228 remain in excited states and emit gamma photons.

Given the following atomic masses :

$-m\left({}^{232}_{90}\text{Th}\right) = 232.0381u$

$-m\left({}^{228}_{88}\text{Ra}\right) = 228.0311u$

$-m\left({}^{228}_{89}\text{AC}\right) = 228.0290u$

$-m\left({}^{4}_{2}\text{He}\right) = 4.0026u$

-1 atomic mass unit $(u) \equiv 931.0\text{MeV}$

Gamma emissions :

-Radium-228: $0.12\text{MeV}, 0.18\text{MeV}$,

-Actinium-228: $0.08\text{MeV}, 0.20\text{MeV}, 0.35\text{MeV}$

A) The kinetic energy of α-particle is 3.73 MeV . (upto two decimal)

B) The maximum kinetic energy of β^{-} particle is 1.95 MeV . (upto two decimal)

C) The total energy (including all gamma rays) released in the process is 6.05 MeV . (upto two decimal)

D) The total energy (including all gamma rays) released in the process is 5.05 MeV . (upto two decimal)

33. Which of the following calculations/results are correctly rounded according to the rules of significant figures?

A) $0.0120 \times 150 = 1.80$

B) $120.0 + 0.076 = 120.076$

C) $\dfrac{6.022 \times 10^{23}}{3.0} = 2.0 \times 10^{23}$

D) $\dfrac{0.00679}{0.134} = 0.050$

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34. A student measures the density of a hollow cylindrical tube using both a screw gauge (for inner and outer diameters) and a vernier caliper (for length). The measurements are as follows :-

Screw Gauge:

Pitch $= 0.5$ mm

. Number of divisions on thimble $= 50$

. For outer diameter: $\text{MSR} = 4.0$ mm, $\text{TSR} = 23$

. For inner diameter: $\text{MSR} = 2.5$ mm, $\text{TSR} = 45$

Zero error $= -0.02$ mm

Vernier Calipers:

. 1 main scale division $= 1$ mm

. 10 vernier divisions $= 9$ mm

. MSR $= 102$ mm

. Vernier coincidence $= 6$ th division

Zero error $= +0.1$ mm

Additional Data:

. Mass of the tube $= 12.34$ g (exactly known)

. Volume of the hollow cylinder $V = \pi L \dfrac{\left(D^2 - d^2\right)}{4}$

where $D =$ outer diameter, $d =$ inner diameter, and $L =$ length.

Take: $\pi = 3.14$

A) The least count of the screw gauge is 0.01 mm and vernier calipers is 0.1 mm

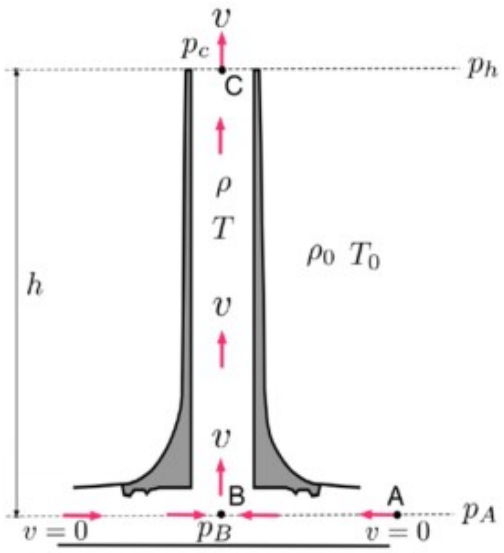
B) The outer diameter (D) is 4.25 mm and inner diameter (d) is 2.97 mm

C) The length (L) using the vernier calipers is 102.5 mm

D) The percentage error in the measurement of density is 1.66%.

35. Solar towers or solar chimneys are structures that make it possible to harness solar energy through air convection. The schematic of a solar chimney can be seen in Figure. At the base of the chimney there are air collectors through which atmospheric air enters the chimney. Solar chimneys are painted black to absorb solar radiation and thereby heat the air inside the chimney. At the same time, the collectors are built from reflective material to heat the air at the base of the chimney. In this way, the air inside the chimney is at a higher temperature than the air outside. The temperature difference causes an airflow from the collectors, at the base of the chimney, toward the upper part of the chimney, where the hot air exits. This airflow can be utilized by placing turbines inside the chimney.

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The purpose of this problem is to understand the operation of a chimney using a simple model. Consider the simplified diagram in Figure. We first describe its operation :

(1) The air outside the chimney is at rest, at ambient temperature T_0, with density ρ_0, and at pressure p_A. This location is marked as (A) in Figure.

(2) Inside the chimney the air is approximately at the same temperature T, higher than T_0 $(T > T_0)$. This temperature difference causes the density of the air inside the chimney, ρ, to be lower than that outside, $\rho < \rho_0$. This in turn generates an air flow which, having velocity $v = 0$ at the entrance, accelerates to a value $v > 0$ at the base of the chimney, marked as (B) in Figure. Very importantly, this also causes the pressure at (B), p_B, to be lower than the pressure at (A), $p_B < p_A$.

(3) This velocity v at (B) remains constant throughout the entire chimney. The air then rises through the chimney at constant velocity v, but when it exits its pressure p_C equals the atmospheric pressure at height h, which we call p_h. A measurement is made of the ambient temperature and of the air temperature inside the chimney, obtaining the values $T_0 = 20^\circ C, T = 50^\circ C$. It is also known that the air density at ambient temperature is $\rho_0 = 1.24\ kg/m^3$. Consider that the height of the chimney is $h = 100\ m$. You may use $g = 10\ m/s^2$.

A) The value of the density of the hot air (ρ) inside the chimney, is $1.12\ kg/m^3$ (approximately).

B) The pressure difference $p_B - p_C$ between points B and C is 1124.8 Pa (approximately).

C) The value of the velocity v with which the hot air exits the chimney is $14.3\ m/s$ (approximately).

D) The value of the velocity v with which the hot air exits the chimney is $20.22\ m/s$ (approximately).

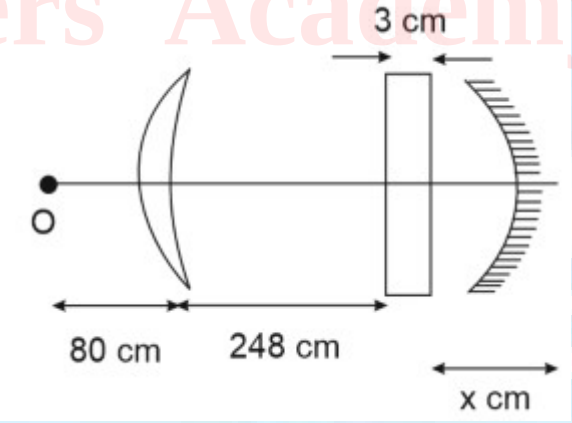
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SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

36. A black plane surface at a constant high temperature T_h is parallel to another black plane surface at constant lower temperature $T\ell$. Between the plates is vacuum. A heat shield consisting of a thin black plate is placed between the warm and the cold surfaces and parallel to these. After some time stationary conditions are obtained. By what factor is the stationary heat flow between the plane surface reduced due to the presence of the heat shield? Neglect end effects due to the finite size of the surfaces.

37. A concavo-convex lens made of glass of refractive index 1.5 has surfaces of radii 20 cm and 60 cm . A concave mirror of radius of curvature 20 cm is placed co-axially to the lens. A glass slab of thickness 3 cm and refractive index 1.5 is placed close to the mirror in the space between the mirror and lens as shown in figure. The distance of the nearest surface of the slab from lens is 248 cm . An object is placed 80 cm to the left of the lens. If final position of the image formed after refraction from lens, refraction from slab, reflection from mirror, refraction from the slab and lens is at the object O , find the distance x (in cm) of the mirror from nearer surface of slab.



38. One end of a steel rod of length ℓ, radius R and modulus of rigidity η is fixed and the other end is twisted through an angle ' θ '. If elastic deformation energy stored in the rod is $k\left(\frac{\pi\eta R^4\theta^2}{32\ell}\right)$, the value of k is :-

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CHEMISTRY **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 12)

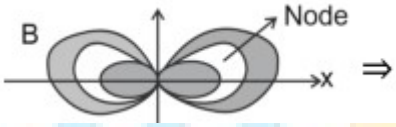
This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

39. Which of the following interhalogen compound have planar, polar and coloured.

A) ClF_5 B) IF_7 C) BrF_3 D) IF_5

40. Which of the following is correct statement regarding orbital/spectra?

A) The degeneracy of second excited state ($n=3$) of H -atom is 9, while the degeneracy of the second excited state of H^- is 3 , irrespective of electron spin being considered.

B) 

$\Rightarrow$ Represents approximate boundary surface diagrams of $3p_x$ for a H-atom.

C) For 1 s electron in a hydrogen atom, the ratio of the probabilities that an electron in a hydrogen atom be at the distances a_0 and $a_{0/2}$ from the nucleus is $2e^{-1}$.

$$\left[\Psi_{1s}(H-\text{atom})=\frac{1}{\sqrt{\pi}a_0^{3/2}}e^{-r/a_0}\right]$$

D) The absorption spectra and emission spectra of a gaseous sample are not necessarily photographic negative of each other.

41. Which of the following statement is correct regarding the test of cation/anion during salt analysis?

A) From the mixture of Fe^{3+} and Al^{3+}, they cannot be separated from each other by using excess NaOH solution

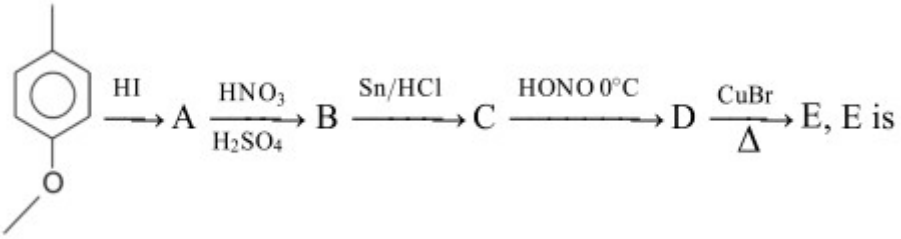
B) When solution of $FeCl_3$ is successively treated with KSCN then colour changes from yellow to red due to redox reaction.

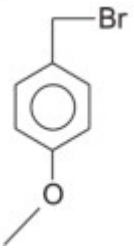
C) When $BiCl_3$ solution is diluted, then white turbidity appears and disappears when dil. HCl is added

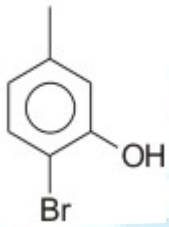
D) SnS is insoluble in YAS

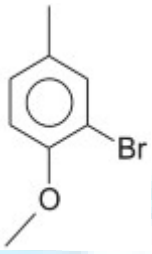
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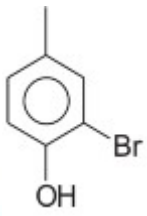
42.



A) 

B) 

C) 

D) 

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 43 and 44:

The molar enthalpy of vaporization of benzene at its boiling point (350 K) is $30.84 kJmol^{-1}$. [Given : R = 8.314 J/K-mol]

43. What is the molar internal energy change (in kJ) ?

44. For how long (in sec) would a 12 volt source need to supply a 0.5 A current in order to vaporise 7.8 g of the sample at its boiling point?

Question Stem for Question Nos. 45 and 46:

Some d-Block elements, form a wide range of coordination compounds with carbonyl and/or phosphines. These complexes have wide range of uses in pharmaceutical synthesis, Agrochemicals, homogenous catalysis, cross-coupling reactions.

Example : Compound (1) : $\left[Pd(PPh_3)_4\right]$,

Compound (2) : $\left[Rh(PPh_3)_3C\ell\right]$,

Compound (3) : $\left[IrC\ell(CO)(PPh_3)_2\right]$,

Compound (4) : $\left[Cu(\eta^5 - C_5H_5)(CO)\right]$,

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Compound (5) : $\left[\left(\eta^5-C_5H_5\right)Ni(NO)\right]$,

Compound (6) : $\left[Mo\left(PF_3\right)_3(CO)_3\right]$,

Compound (7) : $\left[Cu\left(\eta^5-C_5H_5\right)\left(PH_3\right)\right]$,

Compound (8) : $\left[Cr(CO)_6\right]$, Compound (9) : $\left[Ni(CO)_4\right]$

Based on above examples answer the following questions?

45. x_1 = Number of mononuclear compounds which are isoelectronic with $\left[Co(CO)_4\right]^-$.

x_2 = Number of mononuclear compounds which are isostructural with $\left[Co(CO)_4\right]^-$.

Find the value of $\left(x_1+x_2\right)$.

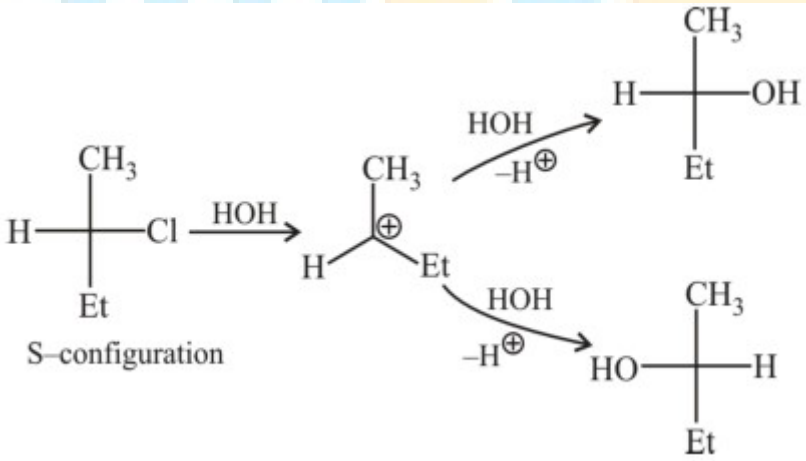
46. x_3 = Number of compounds following Sidgwick EAN rule.

x_4 = Number of compounds where synergic bonding takes places between the metal centre and the anti-bonding orbital of ligand.

Find the value of $\left(x_3+x_4\right)$.

Question Stem for Question Nos. 47 and 48:

S_N2 mechanism also runs along with S_N1 mechanism in small amounts.



47. Calculate the % of R configuration in product, if racemisation is 98% in above reaction.

48. Calculate the % of S_N2 mechanism, if ethyl group is replaced by phenyl group in above reactant and S-configuration in product was 49.4%.

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SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

49. Supercooled rain water $(-1^{\circ}C)$ strikes the wind screen of a car and converts to ice in winter season. With reference to second law of thermodynamics which statement is correct for above process.

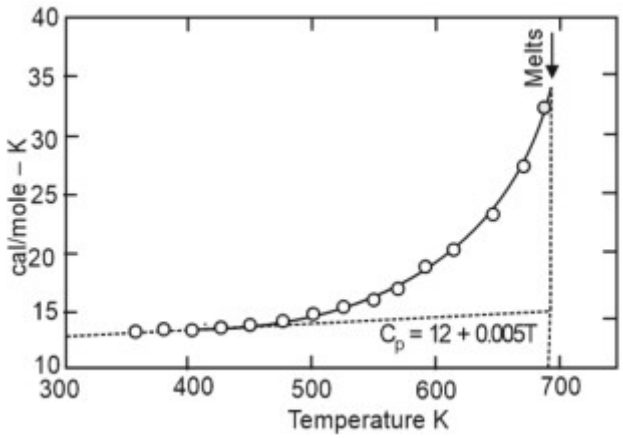
A) Randomness of universe decreases as ice possess less disorder than liquid water.

B) Randomness of surrounding decreases.

C) Water to ice conversion is equilibrium process thus increase in randomness is surrounding is equal to decrease in randomness of system.

D) Increase in randomness of surrounding is greater than decrease in randomness of system.

50.



Graph: Temperature variation of heat capacity of silver bromide.

Which of the following is (are) correct statement(s) for Ionic solids?

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A) In pure alkali halides the most common lattice vacancies are Schottky defects; in pure silver halides the most common vacancies are Frenkel defects.

B) In AgBr , the thermal generation of lattice defects gives an extra contribution to heat capacity of crystal due to their energy of formation.

C) Addition of $CdBr_2$ as impurity in AgBr leads to increase in electrical conductivity and increase in density of crystal.

D) Frenkel defects start appearing appreciably when the Gibb's free energy change becomes favourable mainly due to increased thermal entropy compensating for the defect formation enthalpy.

51. Choose the correct options-

A) Minimum distance between two cubical void in simple cubic unit cell is a . (a = edge length)

B) Crystalline solids are isotropic in nature

C) NaCl structure contains effectively $4'Na^{+}'$ and $4'Cl^{-}'$ ion per unit cell

D) ZnS show frenkel defect

52. Attractive intermolecular forces are known as van der Waals forces, in honour of Dutch scientist Johannes van der Waals (1837-1923), who explained the deviation of real gases from the ideal behaviour through these forces. Van der Waals forces vary considerably in magnitude and include dispersion forces or London forces, dipole-dipole forces, and dipole-induced dipole forces.

Which of the following is(are) correct statement(s) regarding the above type of forces?

A) Generally head to tail colinear orientation is favoured for the nearly spherical molecules while the antiparallel orientation is favoured for the slender dipole molecules.

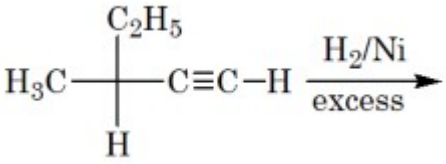
B) $C_2F_6 > C_2H_6$: boiling point order

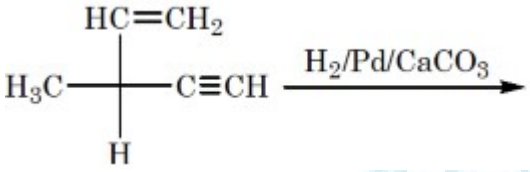
C) The energy associated in interaction between HF and $H_2O \propto r^{-3}$

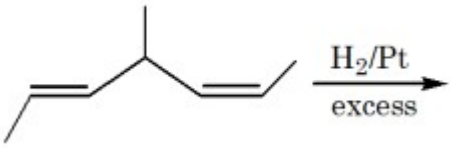
D) For nonpolar molecules, on the time average molecule is polar and at a particular instant, the molecule is non polar.

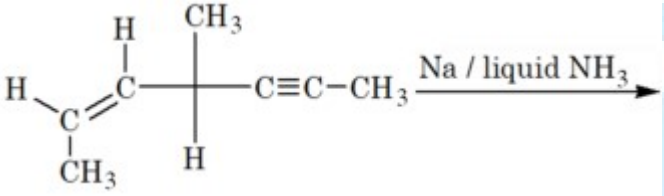
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53. In which of the following reactions the product is optically inactive :-

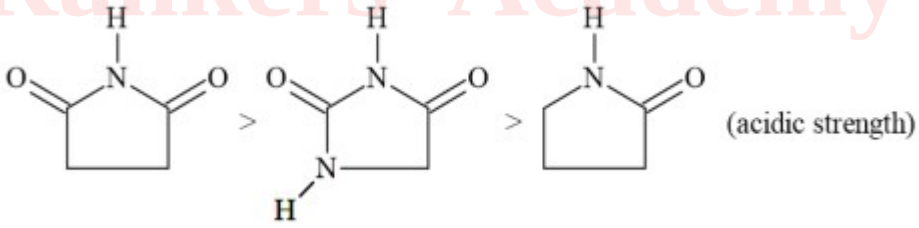
A) 

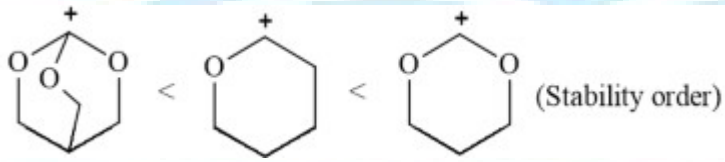
B) 

C) 

D) 

54. Which of the following statements is/are correct?

A) 

B) 

C) : CH_2 (singlet) $>$: CH_2 (triplet) (stability order)

D) $(CH_3)_2NH > CH_3-NH_2 > (CH_3)_3N > NH_3$ (Basic strength in aqueous solvent)

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SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

55. A fuel cell develops an electric potential from the combustion of octane at

1 bar & 298 K

$$C_8H_{18(g)} + \frac{25}{2}O_{2(g)} \rightarrow 8CO_{2(g)} + 9H_2O_{(\ell)}, \Delta G^0 = -3860 \text{ kJ}$$

The E°_{cell} of the fuel cell (in mV) is ? $[1\ F = 96500C]$

56. $XeO_3 + XeOF_4 \rightarrow$ only product (A)

Total number of d-orbital use in bonding in product (A) is x

Total number of $p\pi - p\pi$ bond present in product (A) is y

then find the value of $(x + y)$

57. The optical rotation of the α-form of a pyranose is $+150.7^0$, that of the β-form is $+52.8^0$. In solution, an equilibrium mixture of these anomers has an optical rotation of $+80.2^0$. The percentage of the α-form in equilibrium mixture is :

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